VEDANTAM REVANTH SAI

AI & Data Engineering | LLM Developer | Full-Stack Engineer
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PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Engineering for AI) student at KL University with a 9.31 CGPA and 1 year of industry experience developing LLM-based applications at Viswam.ai. Skilled in building end-to-end AI pipelines — from data ingestion and model fine-tuning to production deployment — using Python, TensorFlow, XGBoost, and the HuggingFace ecosystem. Delivered 7 production-grade projects spanning public health AI, fraud detection, healthcare management, hospital systems, and intelligent scheduling.

EXPERIENCE

AI Intern — LLM Developer | **Viswam.ai** Jan 2024 – Dec 2024

- Developed and fine-tuned LLM-based applications using HuggingFace Transformers and Python; improved model response relevance by ~30% across 3 production deployments through prompt engineering and context management.
- Designed end-to-end text preprocessing and data handling pipelines, reducing data preparation time by ~40% and enabling faster training cycles.
- Integrated REST APIs and backend services to ship AI-powered features; collaborated with a cross-functional team of 6 engineers across design, backend, and QA.
- Applied BLEU, ROUGE, and latency benchmarks to assess model outputs; implemented batching and caching optimizations reducing average inference latency by 25%.
- Architected intelligent systems end-to-end — data ingestion → preprocessing → model training → API deployment — using Python, HuggingFace, Flask, and Docker.

PROJECTS

ProCare — Hospital Management System 2026

Spring Boot · React.js · JWT · PostgreSQL · iTextPDF · JavaMailSender · Swagger · Docker

- Built a full-stack hospital management platform with Spring Boot backend and React.js frontend; implemented JWT-based authentication with role-based access control for admin, doctor, and patient roles.
- Developed modules for patient records, appointment scheduling, and prescription management with RESTful APIs documented via Swagger/OpenAPI.
- Integrated iTextPDF for automated medical report generation and JavaMailSender for email notifications (appointment reminders, prescriptions, discharge summaries).
- Containerised the application with Docker for consistent deployment; designed relational data model in PostgreSQL optimised for healthcare workflows.

GigShield — AI Fraud Detection Platform 2025

React.js · Node.js · Express.js · MongoDB · Python · Scikit-learn · JWT · Stripe · Docker · AWS

- Built an AI-powered trust and security platform for the gig economy; trained anomaly detection and user-behaviour classification models achieving ~92% precision on a 50K-record synthetic dataset.
- Implemented JWT + bcrypt authentication, role-based dashboards, and WebSocket real-time notifications; integrated Stripe and Razorpay APIs for payment monitoring.
- Deployed full MERN stack with a Python ML microservice on AWS/Vercel; achieved <200 ms average API response time under load testing.

Disease Outbreak Detection System 2025

Python · TensorFlow · XGBoost · Flask · React.js · GeoPandas · Folium · Docker · AWS · MongoDB · PostgreSQL

- Built a public health surveillance system with a multi-model ML pipeline combining LSTM time-series forecasting, XGBoost classification, and Isolation Forest anomaly detection — achieving 88% accuracy on historical WHO/CDC datasets.
- Integrated GeoPandas and Folium for interactive geographic risk-zone heatmaps; real-time Plotly/React.js dashboard with automated alerts for 10+ disease categories.
- System processes ~10K daily health records with automated ingestion from public REST APIs via Flask/FastAPI backend.

AI Timetable Generator
Nov 2025 – Feb 2026

Python · Genetic Algorithms · Constraint Satisfaction · Machine Learning

- Engineered a constraint-based scheduling system using Genetic Algorithms, Hill Climbing, and CSP solvers; reduced timetable conflicts by 95% vs. manual scheduling across 3 departments in pilot testing.
- Integrated ML models trained on historical scheduling data to auto-suggest optimal time-slot allocations, cutting manual effort by an estimated 70%.
- Handles multi-department, multi-faculty constraints with configurable rule sets supporting 50+ rooms and 100+ faculty members.

Health Management System
Jan – Dec 2024

Python · Django · PostgreSQL · Heroku

- Full-stack healthcare web application supporting patient registration, appointment scheduling, and medical history tracking with role-based access control for doctor/patient roles.
- Deployed on Heroku with PostgreSQL; optimised query performance with indexing, reducing average page load time by 35%.

Job Portal with LinkedIn Integration
Jan – Dec 2024

Spring Boot · Thymeleaf · LinkedIn API

- Dual-sided job platform for seekers and employers with smart filtering, resume management, and LinkedIn OAuth integration; reduced new user onboarding time by ~60%.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, SQL
AI / ML	Scikit-learn, TensorFlow, Keras, XGBoost, Pandas, NumPy, LangChain, RAG Pipelines, Prompt Engineering, LLM Fine-tuning, Anomaly Detection, Time Series Forecasting, Vector Databases (FAISS, ChromaDB)
Web Frameworks	Django, Flask, FastAPI, Spring Boot, React.js, Node.js, Express.js, Thymeleaf
Databases	PostgreSQL, MongoDB, MySQL, Firebase
DevOps / Cloud	Git, GitHub, Docker, CI/CD (GitHub Actions), AWS, Google Cloud, Heroku, Vercel, Postman, iTextPDF, JavaMailSender, Swagger
Data Viz & GIS	GeoPandas, Folium, Plotly, Matplotlib, Seaborn, Google Maps API
CS Fundamentals	Data Structures & Algorithms, Computer Networks, DBMS, Software Engineering, MLOps

EDUCATION

B.Tech — Computer Science & Engineering
Aug 2023 – May 2027

KL University, Guntur | Specialisation: Data Engineering for AI | CGPA: 9.31 / 10

Relevant Coursework: Data Structures, Algorithms, Computer Networks, DBMS, Software Engineering, AI & Machine Learning, MLOps

CERTIFICATIONS

- MongoDB Associate Developer — Node.js

ACTIVITIES & INTERESTS

- Organised 5+ tech meetups, hackathons, and workshops at KL University for 100+ participants.
- Competitive programming — 150+ problems solved on LeetCode and CodeChef across data structures, algorithms, and dynamic programming.
- Open-source contributor — actively maintain personal AI/ML projects on GitHub with documented READMEs and issue tracking.